

IS THE HOT HAND REAL? EVIDENCE FROM A PERMUTATION AND  
HIERARCHICAL-BASED ANALYSIS

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## ABSTRACT

### Is the Hot Hand Real? Evidence from a Permutation and Hierarchical-Based Analysis

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The hot-hand phenomenon, often described as the tendency for individuals to experience prolonged streaks of success that exceed what would be expected under random performance, has been widely studied across many disciplines, particularly basketball. Early studies attempted to evaluate this effect through various techniques, often concluding that the hot-hand phenomenon was largely a myth. However, recent studies have begun to revisit previous analyses using improved statistical techniques, with some claiming evidence of a discernible hot-hand effect. This study examines the presence of the hot-hand effect in the modern NBA by testing whether observed shooting patterns deviate from those simulated under independence. A permutation-based analysis, which preserves observed shot outcomes while randomizing order, reveals that the observed shot sequences generally do not exhibit elevated streak behavior, providing limited evidence of a hot-hand effect beyond what would be expected by chance. To further account for player and season-level heterogeneity, a logistic hierarchical model was constructed to control for in-game variables that contribute to shot success. The estimated effects of being in a "hot" state are small and reinforce the findings of the permutation test after adjusting for additional covariates.

Keywords: Hot hand, Streak shooting, NBA, Permutation test, Logistic hierarchical model.

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## Chapter 1

### INTRODUCTION

The concept of the hot hand in basketball has been fiercely debated among audiences spanning many decades. Fans and players often describe the phenomenon as reaching an elevated state of play following a streak of recent success. One general notion of hot-hand shooting posits that players tend to make a higher percentage of shots relative to their baseline shooting ability following a recent streak of made shots. Proponents of the hot hand argue that the physical and psychological benefits of witnessing a streak of success place players “in rhythm”, a state of increased confidence and ability that may increase the chances of making a shot. Former Golden State Warriors player Purvis Short described the experience of being “hot” as follows: “You’re in a world all your own. It’s hard to describe. But the basket seems to be so wide. No matter what you do, you know the ball is going in” (Tversky & Gilovich, 1989). However, opponents argue that these experiences may reflect cognitive biases, suggesting that apparent streaks may arise purely from random variation. These perspectives highlight the intuitive appeal of the hot-hand phenomenon, raising the question of whether perceived streaks in performance reflect underlying changes in ability under unique circumstances or simply arise from chance.

## Chapter 2

### BACKGROUND

#### 2.1 Previous Literature

Statistical research has questioned whether the hot hand truly exists, despite widespread belief from fans, players, and the general public. In a seminal study, Gilovich, Vallone, and Tversky (1985), examined streak-shooting tendencies to determine if observed streaks exceeded those expected under randomness. The sample data consisted of field goal attempts from 48 home games of 76ers players and their opponents during the 1980-81 season. The first section of their analysis used conditional proportions to examine players' shooting percentages conditional on the outcome of previous shots. For each of nine players on the 76ers, Gilovich, Vallone, and Tversky computed the overall shooting proportion as well as conditional shooting proportions based on whether the previous one, two, or three shots were successful. They also estimated the serial correlation between the outcomes of successive shots to determine if there existed any dependence structure within a player's shot sequence. A "success", or "hit", was defined as a made shot. The authors discovered that only one out of the nine sampled players possessed a higher success rate on shots following a make compared to their overall success rate. Furthermore, eight out of nine players exhibited a negative serial correlation value, none of which being significantly different from zero.

The findings in this paper contributed to the characterization of the hot hand as a myth, which researchers attribute to a cognitive fallacy whereby individuals misinterpret random streaks as performance trends (Gilovich, Vallone, & Tversky, 1985). A potential explanation for such a phenomenon involves memory bias, the idea that observers are likely to overestimate the dependence of sequential shots due to long sequences of successes (or failures) being more memorable than alternating successes and failures (Gilovich et al., 1985). Other studies have

148 since expanded upon the analysis of Gilovich, Vallone, and Tversky, yielding similar results  
149 of a lack of hot-hand effect (Koehler & Conley, 2003; Lantis & Nesson, 2021; McNair,  
150 Margolin, Law, & Ritov, 2020).

151 However, recent research has revisited the validity of the methods used in Gilovich, Vallone,  
152 and Tversky (1985). In particular, Miller and Sanjurjo (2018) identified a flaw in the  
153 conditional proportion analysis used in the 1985 paper. The flaw stems from a systematic  
154 bias in the analysis of conditional shooting percentages within finite shot sequences, which  
155 ultimately resulted in an underestimation of the true hot hand effect. Miller & Sanjurjo  
156 (2018) formally demonstrated the downward bias in conditional proportion estimators by  
157 analyzing all possible permutations of short finite sequences of independent binary outcomes.  
158 In their example using three consecutive coin flips, they examine the proportion of heads  
159 following a previous head under the assumption of independence. A calculation of the  
160 conditional proportion estimator revealed that the expected proportion of heads conditioned  
161 on a previous head is equal to  $\frac{5}{12}$ , notably lower than the suspected 0.5, illustrating the  
162 systematic downward bias introduced by conditioning on finite sequences (Miller & Sanjurjo,  
163 2018).

164 Miller and Sanjurjo applied the coin-flipping framework to demonstrate bias in the method-  
165 ology of Gilovich, Vallone, and Tversky (1985), whereby the conditional proportions sys-  
166 tematically underestimated the true hot hand effect of the data. Coin flips that resulted in  
167 heads were treated as made shots, while coin flips that resulted in tails were treated as missed  
168 shots. Applying the same logic to a basketball context, Miller and Sanjurjo reasoned that  
169 finite samples of shot data exhibited the same bias observed in the coin problem. More  
170 specifically, the expected proportion of made shots conditioned on a previous make was  
171 not equal to the player's overall shooting percentage, but instead was biased downwards  
172 due to the structure of the sampling process. To fix this issue, Miller and Sanjurjo utilized a  
173 permutation test, described in detail in Section 4.1. After accounting for the downward bias in

conditional proportions with the permutation test, Miller and Sanjurjo discovered a reversal of Gilovich's findings, indicating a discernible hot-hand effect for the given data.

## 2.2 Contextual Variation in Hot-Hand Evidence

While much research has been conducted on hot-hand trends, the settings of each analysis provide little overlap and, therefore, are applied in different contexts. In addition to Gilovich's analysis of 76ers players during the 1980-81 season, an analysis of conditional proportions under a controlled shooting experiment with the Cornell intercollegiate men's and women's basketball teams revealed that the outcomes of shot attempts were largely independent of the outcome of the previous attempt (Gilovich et al., 1985). The controlled experiment required players to shoot under highly regulated conditions, along with monetary incentivization determined by the accuracy of their shots. These results were ultimately subject to the same downward bias introduced by Miller and Sanjurjo.

Other analyses have yielded different results utilizing alternative methods and settings. An analysis of serial dependence on 110,513 free throws in the NBA revealed a discernible, albeit small, hot-hand effect in free throws (Mews & Ötting, 2023). This analysis accounted for unevenly spaced time points between free throws by using a state-space model that assesses a player's form using the Ornstein-Uhlenbeck process. Data from the NBA Three-Point Contest also provide evidence of a hot-hand effect using the permutation testing procedure presented by Miller and Sanjurjo (Ross, 2017). The aforementioned analyses attempt to isolate the hot-hand effect by considering shots under controlled conditions to prevent additional covariates from influencing the results. While this approach yields strong results, they may not generalize to real game situations.

Additional research has offered unique perspectives on hot-hand trends by incorporating context into a player's shots. An analysis of over 83,000 real game shots from the 2012-13 NBA season revealed that players shooting over their baseline shooting percentage in

199 recent shots face different circumstances compared to when they are in their baseline state  
200 (Bocskosky, Ezekowitz, & Stein, 2014). These circumstances include longer shot distances,  
201 facing tighter defense, increased likelihood of taking their team’s next shot, and overall  
202 attempting more difficult shots. The same study also determined that players in a “hot” state  
203 are more likely to make their next shot after controlling for shot difficulty.

204 Basketball philosophy has also evolved substantially since Gilovich’s analysis in 1985, when  
205 offensive play was largely characterized by interior scoring. As player skills have evolved,  
206 teams have increasingly emphasized perimeter play and distance shooting (Wang & Zheng,  
207 2022). In addition, factors such as pace of play may further influence shooting patterns.  
208 Together, these changes in offensive strategy and shot selection suggest that conclusions  
209 drawn from earlier periods may not fully generalize to the context of modern basketball,  
210 potentially altering the interpretation of the hot-hand effect.

211 Ultimately, the presence of the hot-hand phenomenon has largely remained unsolved due  
212 to differences in empirical research and statistical examinations of simulated data, which  
213 often rely on distinct assumptions and methodologies (Bar-Eli, Avugos, & Raab, 2006).  
214 Additionally, no clear definition of the hot hand exists, with studies varying in how they  
215 define and measure streak-shooting behavior. These discrepancies have contributed to  
216 ongoing debate regarding the validity of the effect.

## 217 **2.3 Motivation**

218 Many hot-hand analyses have been conducted in controlled settings or within specific game  
219 contexts, such as the NBA Three-Point Contest (Lantis & Nesson, 2024; Ross, 2017) or  
220 free throw shooting (Arkes, 2010; Chang, 2019; Lantis & Nesson, 2021). Many of these  
221 foundational studies were also conducted in a different era of the NBA than in Gilovich,  
222 Vallone, and Tversky (1985), when the style of play differed substantially from today’s game.  
223 These environments offer valuable insights in their respective contexts, but may differ in

224 applicability to modern in-game playstyles.

225 The NBA has undergone notable changes in offensive philosophy from past eras, with a  
226 greater emphasis on perimeter shooting, increased pace of play, and the development of  
227 more versatile offensive skill sets. As a result, modern gameplay reflects a substantially  
228 different environment than previous eras. Therefore, reexamining the hot hand using modern,  
229 in-game data provides a new perspective on the presence and magnitude of streak shooting.  
230 Specifically, this study examines the hot-hand effect across players and seasons in the modern  
231 NBA, with the goal of evaluating streak-shooting tendencies over the past 10 seasons.

### 3.1 Modern NBA Trends

All data and figures presented in this section are derived from publicly available NBA statistics obtained from Basketball Reference (“NBA League Averages - Per Game,” n.d.).

The most notable example of the evolution of offensive strategy is Stephen Curry’s 2015-16 unanimous MVP season, in which he attempted a then-record 886 three-pointers. Figure 3.1 below displays an increasing trend in the percentage of total field goal attempts (FGA) that were three-point attempts (3PA) after the 2015-16 regular season. The season variable is encoded as the ending year of each NBA season (e.g., 2016 corresponds to the 2015-16 season).

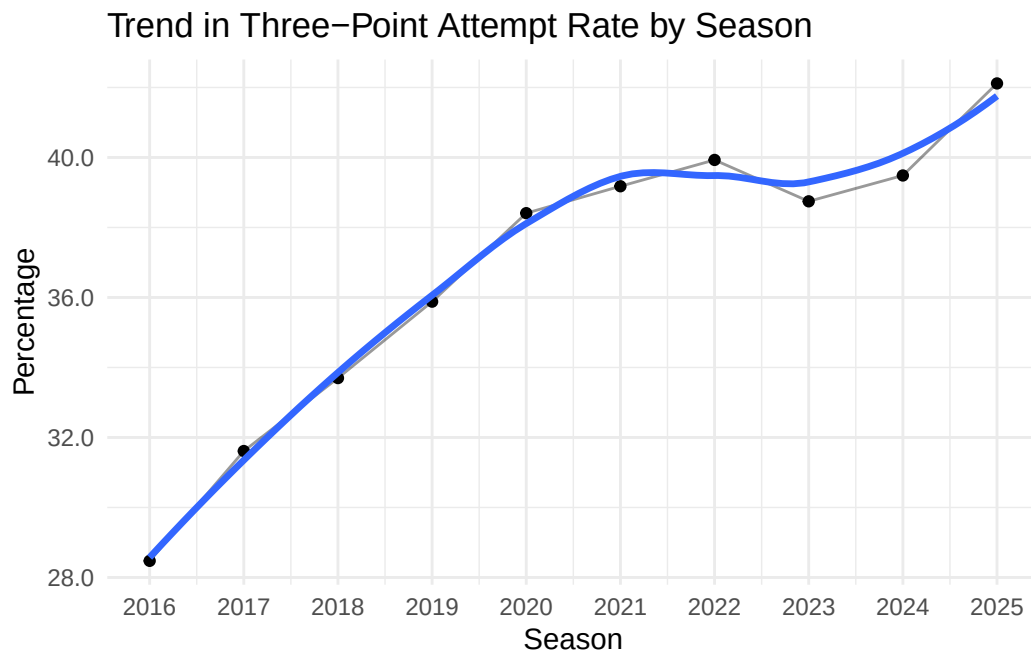


Figure 3.1: Increasing trend in three-point share by season.

243 The increase in three-point shooting raises the question of whether the hot-hand effect has  
244 evolved over time. Improvements in offensive skill could amplify streakiness, whereas the  
245 lower success rate of three-point attempts relative to two-point attempts may attenuate such  
246 effects.

247 Since the 2015-16 regular season, teams have also emphasized a faster pace of play to get  
248 more offensive possessions in a game. Pace, defined as the number of possessions a team  
249 uses in a standard 48-minute game, heavily influences the number of shots taken within  
250 a game and could have an impact on streak shooting. Figure 3.2 below depicts a marked  
251 increase in the league-average pace from the 2015-16 to 2019-20 regular season, followed  
252 by a gradual decline and stabilization in recent seasons.

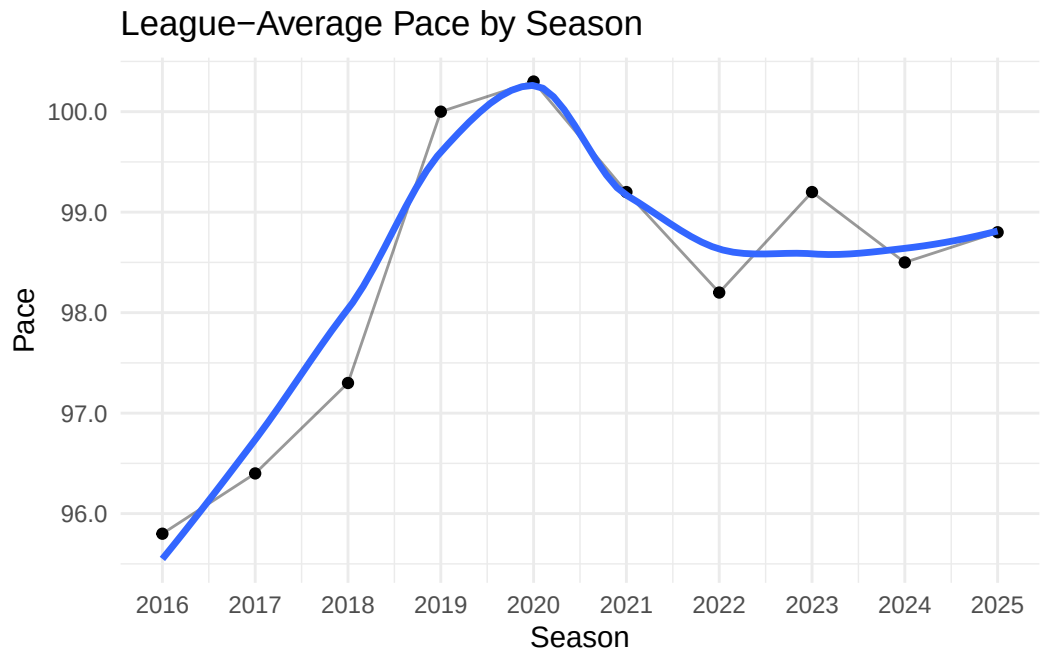


Figure 3.2: League-average pace varies by season, experiencing rapid change after 2016.

253 The fluctuations in pace since 2016 may suggest changes in streak-shooting tendencies  
254 over time, as players attempt more shots per game and therefore have more opportunities to  
255 develop scoring streaks.

256 Figure 3.3 displays the league-average field goal attempts (FGA) per game at the team level



257 increasing at a commensurate rate to the pace of play from approximately the 2015-16 to  
 258 the 2019-20 regular season. Field goal attempts are defined as “any live-ball throw, shot, or  
 259 tap—including layups and dunks—directed at a player’s own basket to score, excluding free  
 260 throws” (*RULE NO. 4*, n.d.).

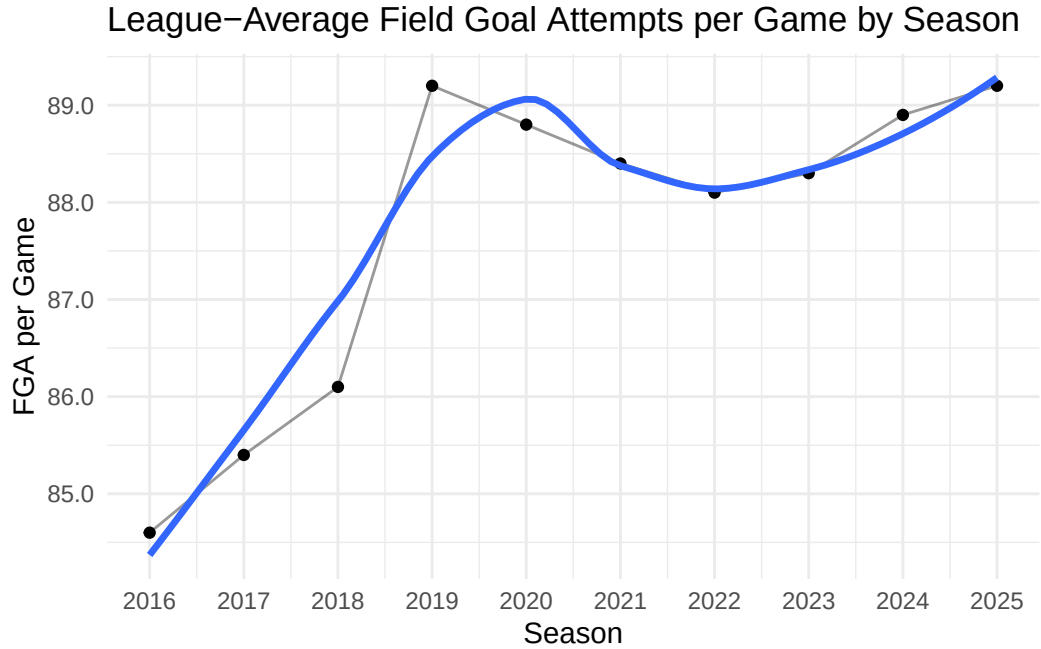


Figure 3.3: League-average FGA per game at the team level increasing after 2016.

261 The increase in field goal attempts per game coincides with the increase of pace, suggesting a  
 262 fundamental alteration in offensive philosophy that may influence streak-shooting tendencies.

### 263 3.2 Data Collection

#### 264 3.2.1 Sample Selection

265 Data on the top 50 players with the most field goal attempts per season were collected for  
 266 each of 10 seasons from 2015-16 to 2024-25 (not including playoffs), yielding a total of 500  
 267 player-seasons. A player-season is defined as a single season of performance for an individual  
 268 player (e.g. 2015-16 Stephen Curry, 2019-20 LeBron James). Restricting the sample to  
 269 the top 50 shooters per season focuses the analysis on high-volume players, who account

for a large share of their team’s shot attempts and are often among the most influential contributors. This approach reduces the overall sample size of the dataset while preserving a high number of shot attempts, allowing for higher statistical power in detecting a hot-hand effect. Importantly, some players were top 50 shooters in FGA in multiple seasons from 2015-16 to 2024-25. These players are represented multiple times in the analysis with unique shot sequences for each respective season in which they were a top 50 shooter. The 2015-16 season was chosen as the starting point due to its significance in changing league-wide trends in shot selection. It is of interest to analyze how the hot hand has progressed since that season, as more teams attempt to replicate the success of prolific offensive teams from that time frame.

All data were collected using the R package *hoopR* (*hoopR Package - RDocumentation*, n.d.), which provides access to NBA play-by-play statistics through SportsDataaverse API. Shot sequence data were obtained for the 500 player-seasons in consideration, with each sequence recording the sequential outcomes of all field goal attempts by a player within a player-season. For example, a shot sequence of ‘0011’, where 0’s represent missed shots and 1’s represent made shots, corresponds to a sequence in which a player missed the first two shots and made the last two shots. A missed shot was also defined as a “failure”, whereas a made shot was defined as a “success”. In total, data from 35,428 games across the 500 player-seasons were collected.

### 3.2.2 Exploratory Data Analysis

The distribution of the total number of field goal attempts per season for the top 50 shooters is displayed in Figure 3.4 below. Notably, the 2019-20 and 2020-21 seasons exhibit lower overall field goal attempts relative to other seasons, which is likely attributable to COVID-19-related restrictions that reduced the number of games played. Prior to those seasons, the distributions of field goal attempts are bimodal, with the larger peaks occurring at approximately 1,000 field goal attempts and the smaller peaks under 1,500 field goal attempts.

After the 2019-20 and 2020-21 seasons, the distributions become more unimodal, with peaks shifting to greater than 1,000 field goal attempts. This pattern could suggest a delayed impact of the 2015-16 season, when long-distance shooting and faster pace became the primary emphasis for most teams. In general, we see an increase in field goal attempts post-COVID, which could alter hot-hand trends.

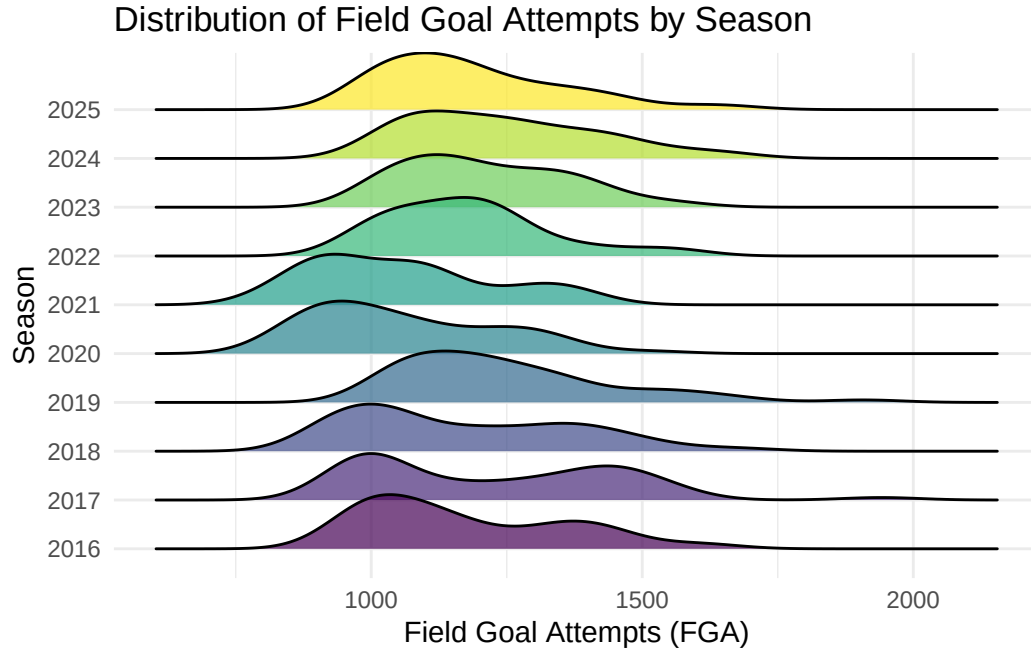


Figure 3.4: Distributions of FGA per game vary by season; shortened seasons in 2020 and 2021.

For each shot sequence, multiple variables were collected or constructed to help analyze the hot-hand effect. An *opportunity* is defined as any shot preceded by at least  $k$  consecutive made shots, and is used as an indicator of whether a player is on a hot streak. In this analysis,  $k = 3$  is used to remain consistent with prior hot-hand literature.

For example, for  $k = 3$ , consider the sequence:

0, 1, 1, 1, 1, 0, 1, 1, 1, 1

Shots that occur immediately after three consecutive made shots are classified as opportunity shots and are underlined in the example above. These opportunity shots are used to evaluate

309 player performance while in a “hot” state.

310 An initial analysis of the data reveals that the mean number of opportunities per game for  
311 qualified player-seasons fluctuates across seasons, albeit at small increments, as shown in  
312 Table 3.1 below. This trend does not necessarily imply any increase in hot-hand shooting  
313 ability, as the increase could stem from a variety of factors. As discussed earlier, league-wide  
314 average field goal attempts per game have also increased over time, which could lead to  
315 a higher number of observed opportunities simply due to a greater volume of shots being  
316 taken.

Table 3.1: Number of Opportunities per Game per Player by Season

| Season | Mean Opportunities |
|--------|--------------------|
| 2016   | 1.12               |
| 2017   | 1.34               |
| 2018   | 1.29               |
| 2019   | 1.36               |
| 2020   | 1.31               |
| 2021   | 1.47               |
| 2022   | 1.36               |
| 2023   | 1.53               |
| 2024   | 1.58               |
| 2025   | 1.42               |

317 To adjust for the increase in the number of field goal attempts taken per game, the percentage  
318 of total field goal attempts that were opportunity shots was calculated and plotted in Figure  
319 3.5. The percentage of opportunity shots shows an overall upward trend across the period, but  
320 with noticeable year-to-year fluctuations. This increasing trend suggests a higher prevalence  
321 of streak occurrences over the past 10 seasons, which could be attributed to changes in

322 offensive strategies or players becoming more capable of entering hot streaks. Importantly,  
 323 this trend only describes a slight increase in the percentage of opportunity shots as opposed  
 324 to an actual hot-hand effect.

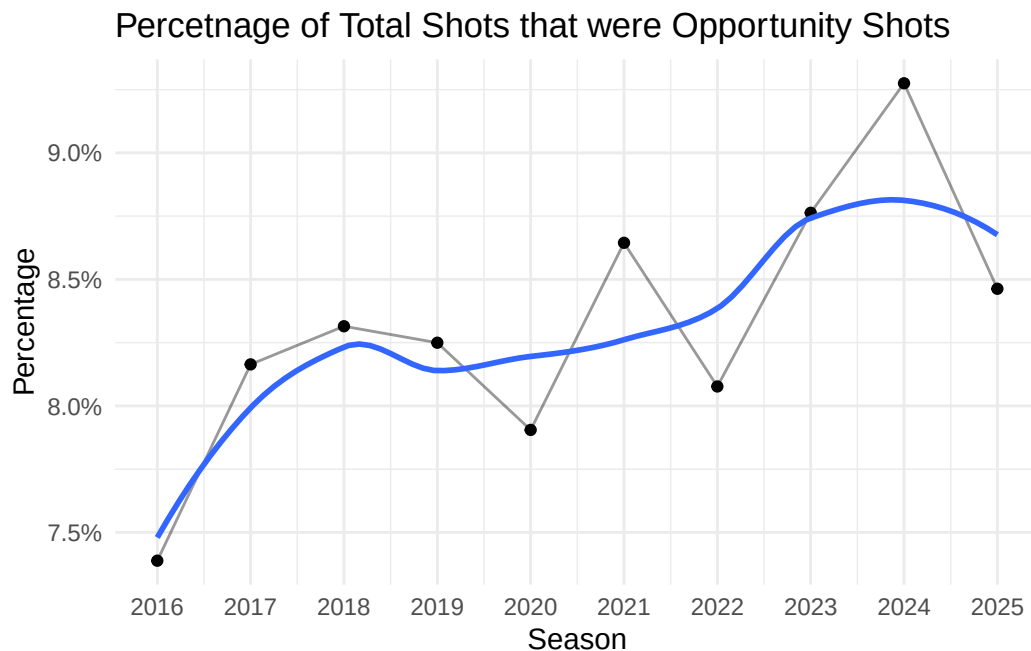


Figure 3.5: Proportion of Opportunity Shots showing general upward trend for qualified player-seasons.

325 Opportunity field goal percentage, defined as the field goal percentage on opportunity shots,  
 326 for qualified player-seasons has also generally increased over time, as shown in Figure 3.6.  
 327 The graph starts at its lowest point during the 2015-16 season and increases steadily until the  
 328 2018-19 season, where it initially peaks. The 2019-20 season displays a steep decline in hot  
 329 hand field goal percentage, but eventually recovers and continues to rise until it peaks during  
 330 the 2023-24 season. The wave-like pattern of the smoothed line suggests variability over  
 331 time rather than a strictly monotonic increase. An upward trend provides some preliminary  
 332 evidence of an increase in hot-hand shooting ability over time, but needs to be formally  
 333 tested for statistical significance as it may be attributable to random variation.

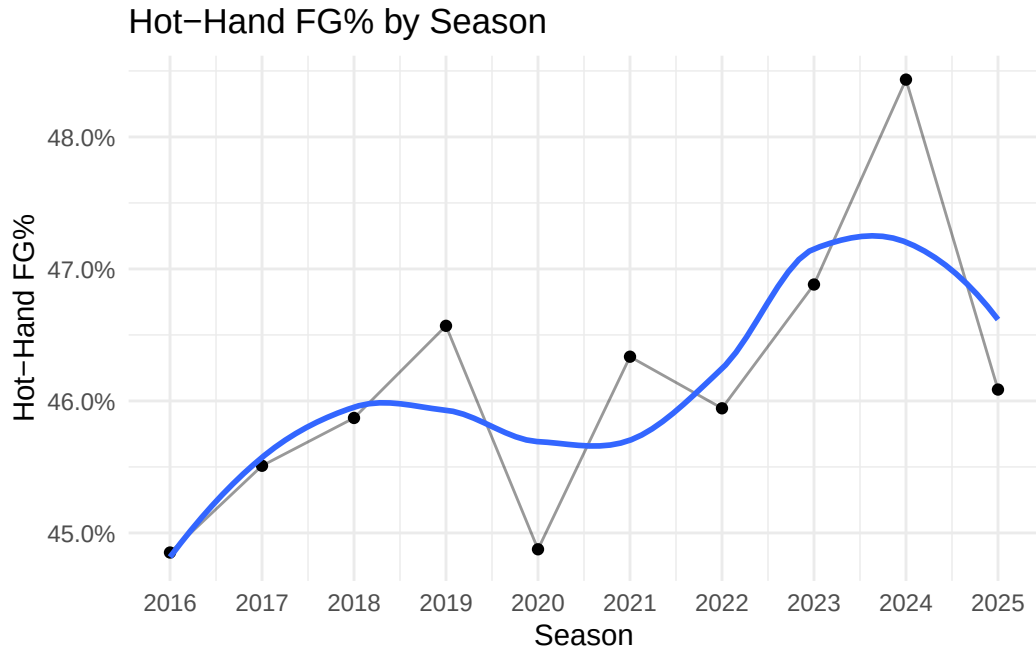


Figure 3.6: Hot-hand FG% showing general upward trend over time for qualified player-seasons.

334 Although the trends discussed above may indicate a higher degree of streakiness, a meaning-  
 335 ful evaluation requires a comparison to baseline shooting performance. Figure 3.7 below  
 336 addresses this by plotting the difference between opportunity FG% and baseline FG% for  
 337 all qualified player-seasons from 2015-16 to 2024-25. For each season, the baseline FG% is  
 338 computed by summing the total number of made field goals from qualified players during  
 339 that season divided by the total number of field goal attempts during that season from quali-  
 340 fied players. This provides a natural benchmark with which the opportunity FG% can be  
 341 evaluated.

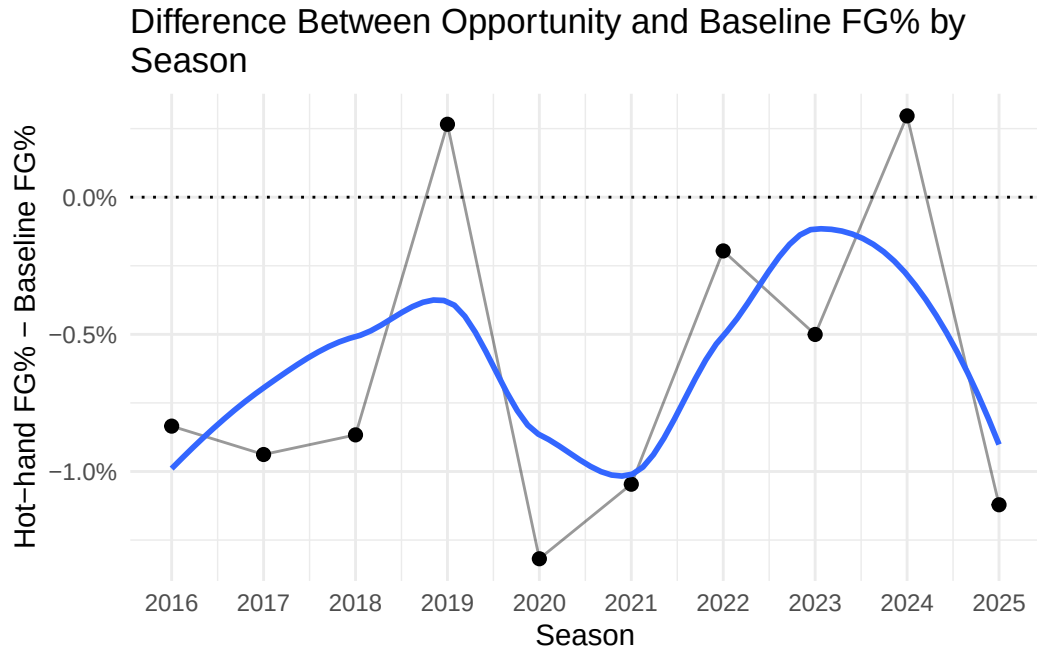


Figure 3.7: Difference between Opportunity and Baseline FG%, with horizontal marker at 0%.

The overall difference seems to fluctuate around zero, with most seasons possessing a slight negative difference, indicating that the success rate on opportunity shots is less than the success rate for baseline attempts on average. While there are isolated seasons, such as 2018-19 and 2023-24, in which hot-hand FG% is greater than the baseline FG%, these deviations are neither persistent nor large in magnitude. These results do not provide any preliminary evidence that a systematic hot hand effect exists on the aggregate level. It is important to note that this analysis remains subject to the selection bias identified by Miller and Sanjurjo, as the hot-hand FG% is computed by conditioning on prior shot outcomes. Therefore, more rigorous and detailed analysis methods that account for this bias are needed before any conclusions can be made regarding the hot-hand effect.

The distance of each shot attempt was collected using hoopR's coordinate system, with each shot containing an x and y-coordinate relative to the basket. Shot distances were then organized into one of three categories: close range (0-8 feet), mid-range (8-22 feet), and

three-pointer (22+ feet). These category labels are approximate, as the distance of a three-point attempt varies depending on court location. For example, corner three-point attempts are taken from approximately 22 feet, whereas three-point attempts taken from the top of the arc must be at least 23.75 feet. The chosen cutoff values reflect the empirical distribution of shot distances, which tend to cluster in specific regions. Figure 3.8 below displays the distribution of shot distances for all games within the 500 qualified player-seasons, with vertical lines marked at cutoff values of 8 and 22 feet.

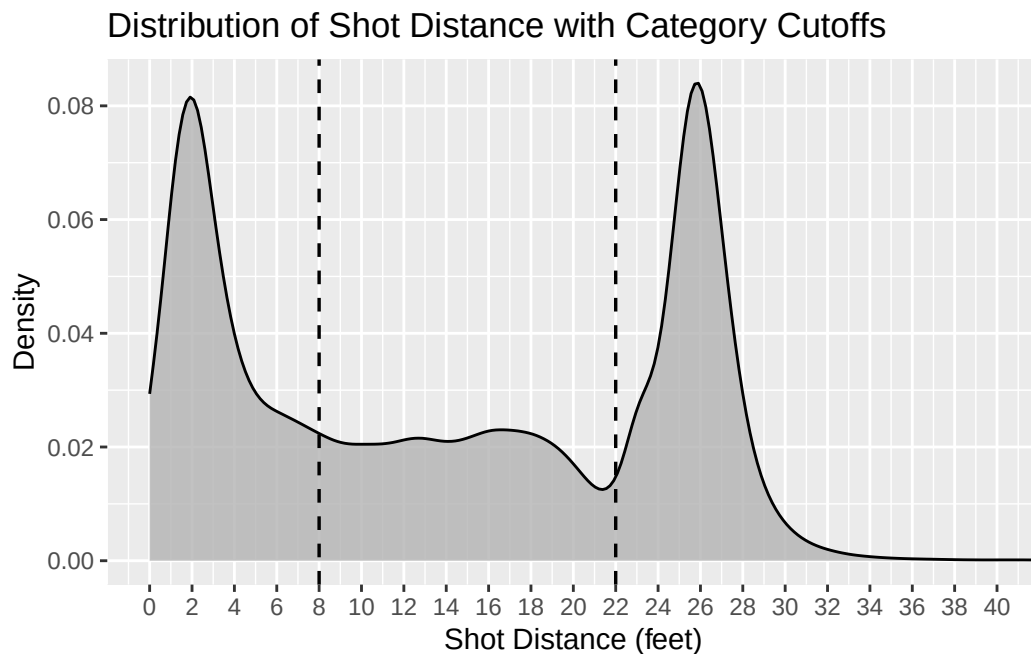


Figure 3.8: Distribution of Shot Distances with cutoff values at 8 and 22 feet.

The density plot exhibits clear clustering of shots near the basket (approximately 0-8 feet) and beyond the three-point line (approximately 22-30 feet), with relatively fewer attempts in the mid-range. This varying structure suggests that teams tend to compartmentalize shot types into various categories rather than a continuous spectrum. Therefore, modeling shot distance as a categorical predictor with three distinct levels provides a more practical and interpretable representation of shot selection than treating it as a purely continuous quantitative variable.



### 3.3 Streak Statistics

Ross (2017) outlines various *streak statistics* that analyze finite sequences of binary outcomes. A streak statistic is a measure designed to quantify the “streakiness” of a given shot sequence. The streak statistic used in my analysis is defined below:

*Difference in proportion of  $S$  (opportunities - non-opportunities)*: The difference between the proportion of success on opportunity shots and the proportion of success on non-opportunity shots. In the previous example, 0, 1, 1, 1, 1, 0, 1, 1, 1, 1, the value of the streak statistic was  $\frac{2}{3} - \frac{6}{7} \approx -0.1905$  (Ross, 2017).

This statistic was chosen because it provides a direct comparison of shooting performance in a “hot” state against shooting performance outside of that state. Additionally, it has an intuitive interpretation under the assumption of independence, as it is often expected to be equal to zero. However, as discussed earlier, this expectation is biased downward in finite samples, which will be revisited in the subsequent analysis.

## Chapter 4

### PERMUTATION TESTING

#### 4.1 Permutation Testing Procedure

The permutation testing procedure utilizes a streak statistic to assess the likelihood of observing a shot sequence at least as extreme as that computed from the observed shot sequence. We define the null and alternative hypotheses:

$H_0$ : Shots are independent.

$H_a$ : Shots are dependent.

The steps of the procedure are outlined below:

1. Calculate the streak statistic for the observed shot sequence.
2. Randomly permute the outcomes of the observed shot sequence while preserving the total number of successes and failures. This produces a sequence consistent with the null hypothesis of independent shot attempts.
3. Calculate the streak statistic for the permuted shot sequence.
4. Repeat steps 2-3 many times (e.g. 10,000 iterations) to approximate the sampling distribution of the streak statistic assuming independence of shot attempts.
5. Calculate the proportion of permuted streak statistics that are at least as extreme as the observed statistic in the direction consistent with the hot-hand effect to obtain the  $p$ -value.

This procedure addresses the bias identified in Gilovich, Vallone, & Tversky (1985) by directly constructing an appropriate null distribution from the observed shot sequence rather

than relying on theoretical expectations under independence. The random permutation of observed outcomes satisfies the null assumption of independence, allowing us to compare the observed shot sequence to many sequences simulated under the null hypothesis.

## **4.2 Season-Level and Within-Game Shot Sequences**

The permutation testing procedure was applied to season-long shot sequences as well as within-game shot sequences to assess the robustness of results across different levels of aggregation. Each approach provides distinct advantages that address limitations inherent in the other.

### **4.2.1 Season-long Data**

Each observation in this sample consists of an individual player's sequential shot data for a particular season. That is, for each of the 500 player-seasons, a binary sequence of makes and misses is constructed using all shots taken over the course of the season. Shots are carried over across games, meaning they accumulate over the course of a season without resetting after each game. As a result, streaks can extend across game boundaries. For instance, if a player makes the final two shots of one game and the first shot of the next game, the subsequent shot would be classified as an opportunity. Treating each game separately, as examined in the within-game analysis, effectively censors shot sequences at the end of each game by truncating potential streaks. This could result in weaker evidence of a hot-hand effect by reducing the number of observed opportunity situations and inflating p-values. By allowing sequences to carry over across games, this analysis avoids inflating p-values and maximizes the statistical power under the given context.

The permutation test is applied to each of these season-long sequences, allowing for the evaluation of streak behavior over a larger sample of shots per player. Using season-long shot data allows for greater statistical power due to the large number of opportunities within a given sequence. With more qualifying observations, estimates of conditional shooting

428 probabilities become more stable and less sensitive to random variation, increasing the  
429 ability to detect meaningful deviations from the null hypothesis. It also offers a unique  
430 perspective on hot-hand shooting, with results applying to prolonged stretches of time rather  
431 than only individual games. Significant results for season-long data could indicate a potential  
432 carry-over effect, with players experiencing increased levels of play across multiple games  
433 given that the player finished the previous game on a streak.

434 Figure \_\_ (full results for all player-seasons provided in Appendix) displays the permutation  
435 test results for the 30 player-seasons exhibiting the strongest evidence of a hot-hand effect.

436 **(TABLE IN APPENDIX)**

437 At an  $\alpha = .05$  individual significance level, only 10 out of 500 player-seasons exhibit  
438 a statistically discernible hot-hand effect. However, when conducting 500 independent  
439 hypothesis tests, we would expect approximately  $0.5 * 500 = 25$  false rejections purely  
440 by chance, even if no true hot-hand effect exists. Thus, the observed number of significant  
441 results is lower than what we would expect under the null hypothesis of independence. Figure  
442 4.1 below displays the distribution of p-values for the 500 player-seasons. The distribution  
443 is skewed left, with many observations possessing p-values greater than 0.05.

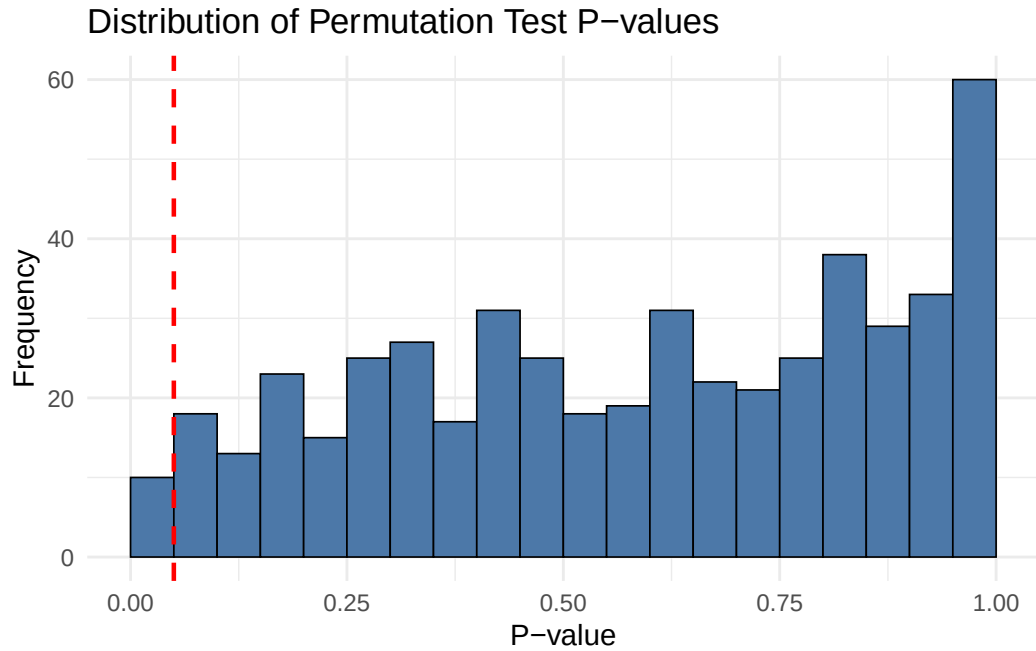


Figure 4.1: Distribution of Permutation Test P-values displaying left skew.

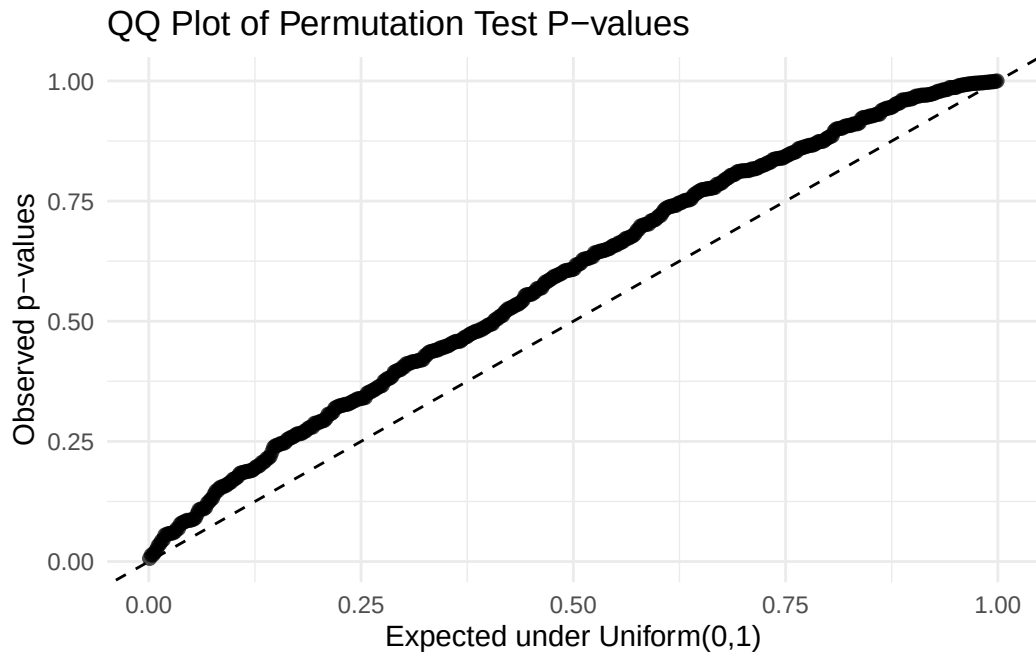


Figure 4.2: Distribution of Permutation Test P-values compared to uniform dsitribution.

444 The QQ Plot (Figure 4.2) above compares the observed p-values from the permutation tests  
 445 to the uniform distribution. Under the null assumption of independence, we expect the

distribution of  $p$ -values to follow a uniform distribution, displayed by the diagonal line. The hot-hand effect suggests that the distribution of observed  $p$ -values should fall below the diagonal line since a larger proportion of player-seasons would be expected to show stronger evidence than under independence, resulting in lower quantiles. However, the observed  $p$ -values align above the diagonal line, which not only suggests that there is not only little evidence of a hot-hand effect, but also that players potentially perform slightly worse during opportunities than other shots. These results are aggregated at the player-season level, based on season-long shot sequences rather than game-level data. As a result, the analysis captures within-season patterns but may mask potential short-term dynamics.

To account for the false discovery rate of hot-hand effects, the Benjamini-Hochberg procedure was applied to the existing  $p$ -values. After this adjustment, none of the player-seasons remain statistically significant, providing no evidence of a hot-hand effect after correcting for multiple comparisons.

#### 4.2.2 Game-level Data

Season-long shot sequences were then deconstructed further into game-level sequences. In this sample, each observation corresponds to a binary sequence of makes and misses for a game within a player-season. Thus, each player-season contributes multiple sequences, one for each game in which the player recorded field goal attempts. The permutation test is applied to each game-level shot sequence separately, capturing within-game streak behavior and allowing for a more granular analysis of hot-hand shooting patterns. Results for game-level data are often considered more practical for hot-hand analysis than season-long data since most people attribute the hot-hand effect to individual games rather than prolonged periods of time. However, using game-level data decreases the statistical power of the permutation test considerably, as shot sequences typically contain very few opportunities, if any.

471 In total, 35,428 shot sequences were considered for the game-level analysis. The distribution  
 472 of total FGA per game for qualified players is displayed below in Table 4.1 and Figure 4.3,  
 473 with a mean of 16.5 FGA per game and a standard deviation of 5.19 FGA per game.

Table 4.1: Total Shots per Game per Player

| Minimum | Q1 | Median | Mean | SD   | Q3 | Maximum |
|---------|----|--------|------|------|----|---------|
| 1       | 13 | 16     | 16.5 | 5.19 | 20 | 50      |

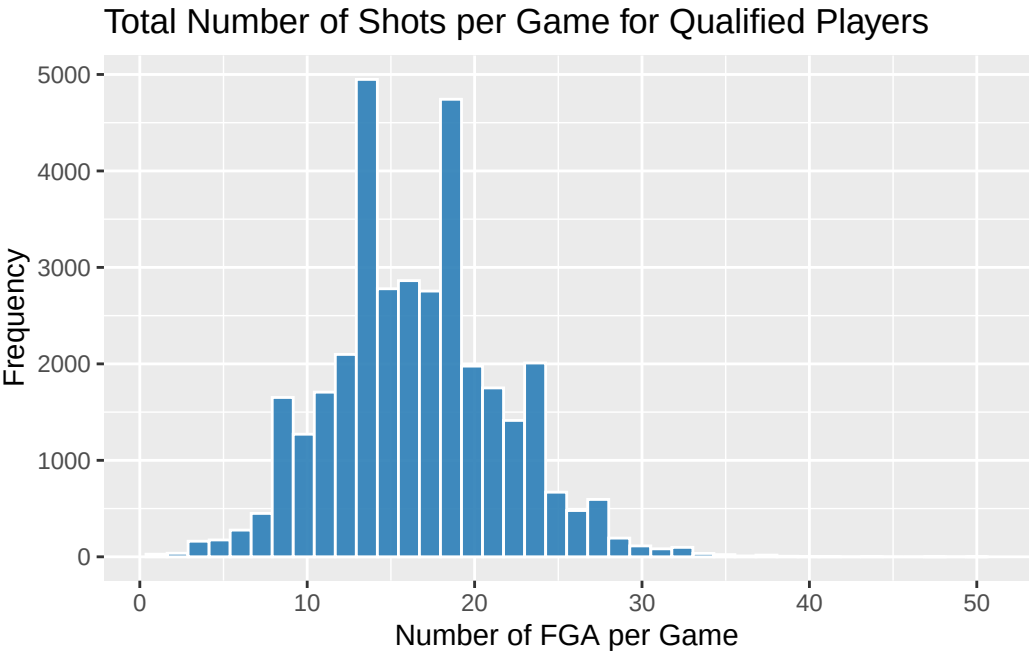


Figure 4.3: Total shots per game for qualified player-seasons.

474 The limited sample size of each shot vector restrains the number of opportunity shots within  
 475 a sequence, thereby reducing the statistical power of the permutation test. In addition, 14,563  
 476 games (approximately 41.1%) in the sample contain no opportunity shots which further  
 477 reduces the effective sample size to 20,865 games. Games with zero opportunity shots were  
 478 retained in the dataset but classified as providing no information for the permutation test.

### 4.3 Quantifying the Bias

Prior work has shown that the magnitude of small-sample bias decreases as sample size increases (Miller & Sanjurjo, 2018). Thus, the magnitude of bias for the season-long data is minimal and has little effect on the observed streak statistic. In contrast, the limited sample size of shot sequences in the within-game analysis warrant a closer examination of this bias. Because the magnitude of the bias is not immediately clear, a simulation-based approach was used to quantify its impact under realistic conditions for the within-game analysis.

### 4.4 MCMC Simulation

A Monte Carlo simulation was used to approximate the expected value of the streak statistic *Difference in proportion of  $S$  (opportunities – non-opportunities)* under different sequence lengths and success probabilities. A streak length of three was used for the simulation. The estimation procedure is summarized in Algorithm 1.

#### INSERT ALGORITHM HERE

Figure 4.4 below displays the expected value of the streak statistic using Monte Carlo simulation. The parameter  $p$  varies from 0.40 to 0.60, in increments of 0.05, reflecting plausible shooting proportions observed in game settings, although more extreme values exist due to small sample sizes. The overall sample proportion of made shots across all 35,428 qualified games was approximately 0.467. For each  $(n, p)$  pair, the simulation was conducted using 10,000 trials and the resulting expected values are presented in the figure below.



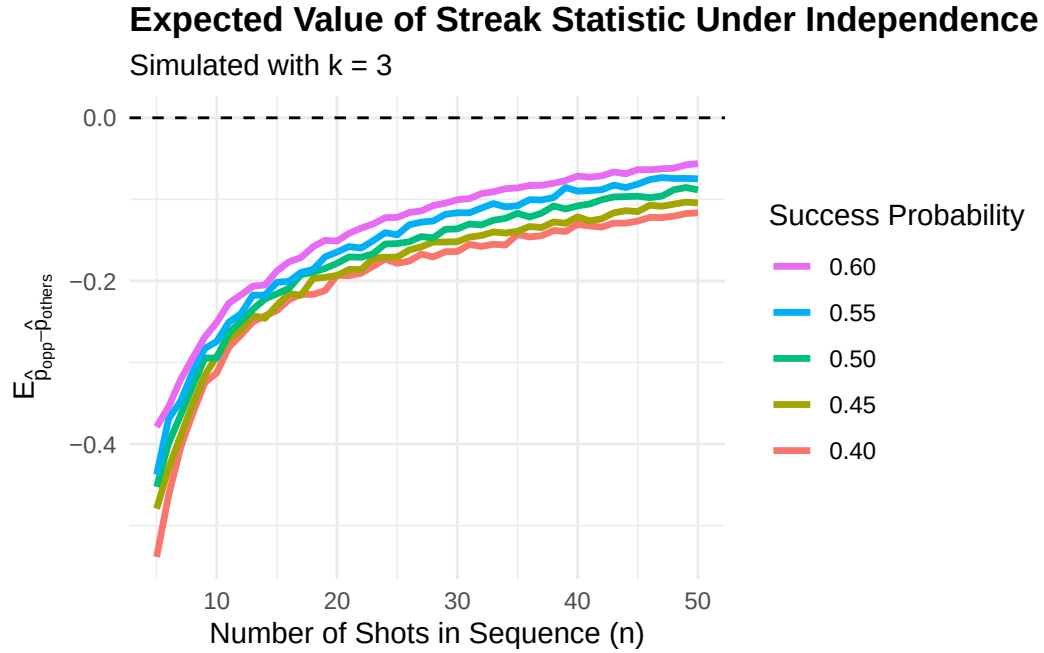


Figure 4.4: Simulated bias of streak statistic under different conditions.

499 The curves exhibit a consistent vertical ordering with higher success probabilities corre-  
500 sponding to a smaller magnitude of bias. Differences in bias across success probabilities  
501 appear relatively consistent, indicating that the bias generally scales proportionally with  $p$ .  
502 Additionally, each line appears to be monotonically increasing towards 0, meaning that the  
503 expected bias decreases as  $n$  increases for a given success probability  $p$ .

504 These results can be compared to the null statistics of each observed shot sequence generated  
505 by the permutation test. However, it is important to note that shot sequences typically vary  
506 with  $p$  and  $n$ , which would alter the magnitude of bias for that particular sequence.

#### 507 4.5 Bias-Adjusted Streak Statistic

508 The permutation test was run for all shot sequences in the sample. Sequences with zero  
509 opportunities were assigned a  $p$ -value of  $NA$ . Table \_\_ in the Appendix contains a sample of  
510 30 games and the collected statistics through the permutation tests. To visualize the effect of  
511 different sample sizes on the null distribution, three games of different sample sizes were

collected. The three selected games contained 13, 16, and 20 shots to represent Q1, the median, and Q3, respectively. Each game had a streak statistic of -0.5 for consistency. The observed and null statistics are shown in Table 4.2 below.

Table 4.2: Bias-Adjusted Streak Statistic for Different Sample Sizes

| Player          | Total Shots | Total Makes | Opportunity % | Difference in Proportions | Null Mean | Bias-Adjusted Difference |
|-----------------|-------------|-------------|---------------|---------------------------|-----------|--------------------------|
| Kyle Kuzma      | 20          | 9           | 0             | -0.5                      | -0.235    | -0.265                   |
| Tobias Harris   | 16          | 7           | 0             | -0.5                      | -0.242    | -0.258                   |
| Jordan Clarkson | 13          | 6           | 0             | -0.5                      | -0.292    | -0.208                   |

The *Difference in Proportions* column represents the streak statistic, defined as the difference between the proportion of makes on opportunity shots and the proportion of makes on all other shots. However, this statistic is subject to the bias discovered by Miller & Sanjurjo (2018). The *Bias-Adjusted Difference* column corrects this statistic by subtracting the null mean obtained from the observed streak statistic. This adjustment accounts for bias in the original streak statistic, yielding a bias-adjusted streak statistic.

As expected, the null means generally seem to decrease as the number of shots increase, although some trend deviation remains due to differences in field goal percentages across games (Figure \_\_, Appendix). The average null mean across all games with at least one opportunity was approximately -0.229, indicating a substantial bias in the streak statistic. This estimated bias agrees with the results obtained from the Monte Carlo simulation, as our overall success probability was approximately 0.467 and the average number of shots taken per game was approximately 16.5. The null standard deviations are relatively large compared

to the season-long analysis, reflecting the high variability inherent in short shot sequences. Because many games contain a limited number of opportunities, the field goal percentage on those opportunities can vary substantially, leading to highly volatile streak statistics. For example, in games with only one opportunity, the opportunity field goal percentage can take on only two values (0 or 1). As a result, the streak statistic is highly discrete, and the corresponding permutation distribution is characterized by a small set of possible values that can differ markedly from one another, contributing to considerable variability in the null distribution. Consequently, games with few opportunities will also have lower statistical power, making it more difficult to detect meaningful hot-hand trends.

#### 4.6 Game-Level Results

Overall, only 562 out of 20,865 games (approximately 2.69%) that included at least one opportunity produced a statistically significant streak statistic at the  $\alpha = 0.05$  significance level. This proportion is lower than what would be expected under the assumption of independence, providing little evidence in support of a within-game hot-hand effect. Furthermore, the mean of the *Adjusted Difference in Proportions* column across all games with at least one opportunity was approximately -0.0136, indicating that, on average, players performed slightly worse on opportunity shots than would be expected under independence.

Figure 4.5 provides the distributions of the bias-adjusted streak statistic by season. The distributions appear largely consistent across seasons, exhibiting similar bimodal shapes and comparable peak locations. In particular, each season shows two prominent peaks centered at approximately -0.25 and 0.25. The black points denote the seasonal means of the null-adjusted streak statistic, which are generally centered slightly below zero. This aligns with the previous finding that players tend to perform marginally worse on opportunity shots relative to other shots after adjusting for the bias. One exception is the 2018-19 season in which the mean appears slightly above zero, suggesting a small, though likely negligible, positive hot-hand effect. The similarity between the distributions suggests little evidence of

554 systematic seasonal differences in hot-hand effects.

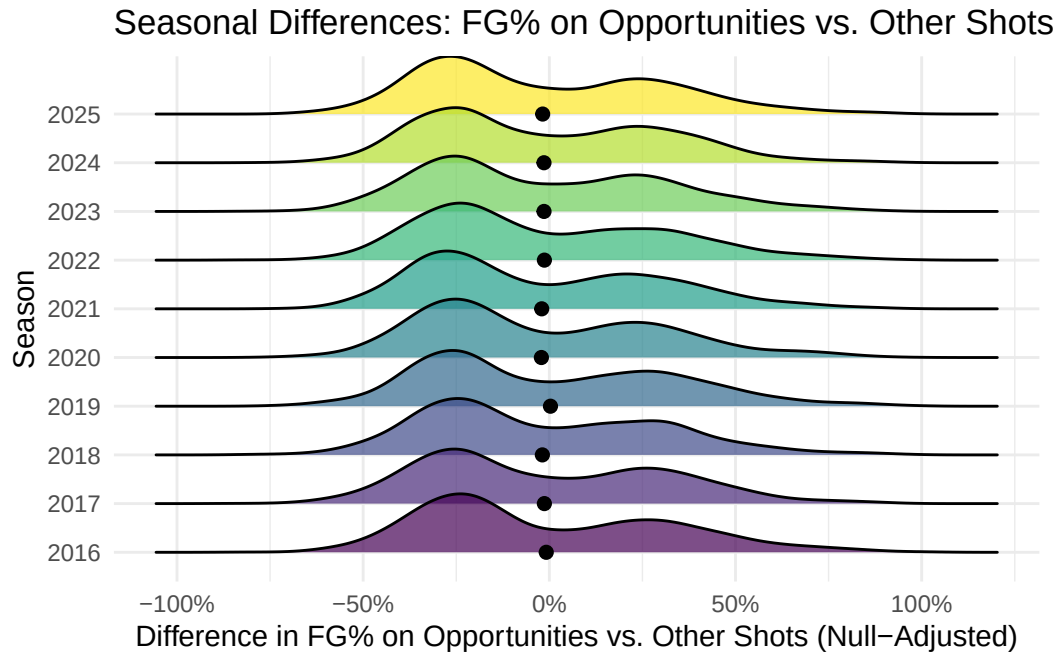


Figure 4.5: Bias-Adjusted Streak Statistic Distributions by Season.

555 The observed bimodality can be explained by the structure of the streak statistic that results  
556 from small shot sequences. Because many games contain few opportunities, the opportunity  
557 field goal percentage is often either very low (e.g., 0) or moderate (e.g., around 0.5), while the  
558 field goal percentage on other shots is typically near 0.5, reflecting both the larger number  
559 of non-opportunity (“other”) shot attempts in each game and the fact that league-wide  
560 field goal percentages tend to be close to this value. Consequently, the unadjusted streak  
561 statistic frequently takes values near -0.5 or 0. Figure 4.6 displays this pattern by using the  
562 three previously sampled games which contained different numbers of total shots ( $n$ ) and  
563 successful shots ( $m$ ). The null distributions reveal points clustering at approximately -0.5  
564 and 0, confirming our previous intuition.

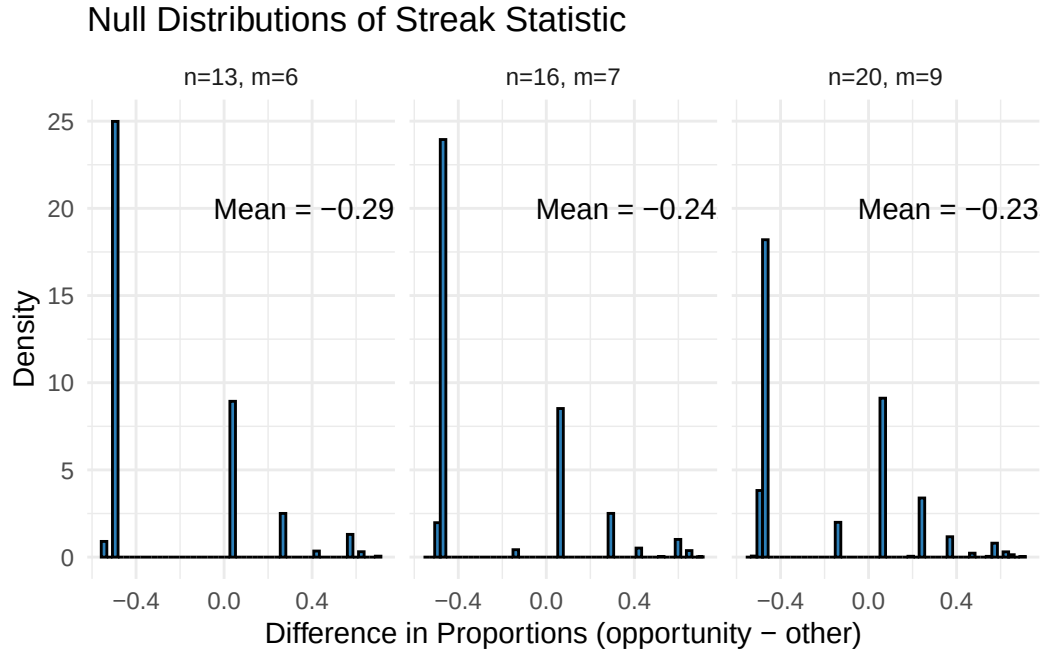


Figure 4.6: Null Distributions of Streak Statistic under Different Scenarios.

After adjusting for the null mean (approximately -0.23), these values shift to around -0.25 and 0.25, producing the observed peaks. This pattern reflects the high variability of the statistic rather than systematic differences across seasons, providing little evidence of a meaningful season-specific hot-hand effect.

To examine the results further, the games were categorized by the number of opportunities in their respective sequence in Table 4.3. The table shows an overall negative relationship between the number of opportunities within a game and the number of games associated with the number of opportunities. This is expected because qualifying for opportunity shots is increasingly difficult as the number of opportunities increases. The *Proportion of Games with Adj\_Diff > 0* column represents the proportion of games in which the bias-adjusted streak statistic was greater than 0, indicating a positive directional effect in favor of the hot hand.

Table 4.3: Proportion of Games with Bias-Adjusted Difference in Proportions Greater than 0.

| Number of Opportunities | Number of Games | Proportion with Positive Adjusted Difference |
|-------------------------|-----------------|----------------------------------------------|
| 0                       | 14563           | NA                                           |
| 1                       | 8184            | 0.043                                        |
| 2                       | 5401            | 0.676                                        |
| 3                       | 3341            | 0.644                                        |
| 4                       | 1912            | 0.763                                        |
| 5                       | 984             | 0.774                                        |
| 6                       | 545             | 0.813                                        |
| 7                       | 282             | 0.791                                        |
| 8                       | 122             | 0.779                                        |
| 9                       | 59              | 0.881                                        |
| 10                      | 16              | 0.688                                        |
| 11                      | 12              | 0.750                                        |
| 12                      | 3               | 0.667                                        |
| 13                      | 3               | 0.333                                        |
| 14                      | 0               | NA                                           |
| 15                      | 1               | 1.000                                        |

577 The proportion of games with adjusted difference in proportions is undefined for games  
 578 with exactly 0 or 14 opportunities. Games with zero opportunities do not yield a streak  
 579 statistic, and zero games in the sample contain exactly 14 opportunities. Among games with  
 580 at least one opportunity, those with only one opportunity comprise approximately (39.22%),  
 581 representing the largest share across all opportunity counts. However, only about 4.3% of  
 582 these games exhibit a positive null-adjusted difference in proportions, indicating minimal

583 evidence of a hot-hand effect for single-opportunity games.

584 This result is consistent with the structural constraints of the shot sequence and qualifications  
585 of an opportunity shot. For a game to contain exactly one opportunity, the player must miss  
586 the opportunity shot. Otherwise, a made opportunity shot would generate an additional  
587 opportunity on the subsequent attempt. Consequently, the only scenario in which a positive  
588 adjusted streak statistic can occur in a one-opportunity game is when the lone opportunity  
589 arises at the end of the sequence, such as when the player makes the final four shots of  
590 the game with no prior opportunities. In this case, the final shot constitutes a successful  
591 opportunity while precluding the possibility of additional opportunities. Because this scenario  
592 occurs infrequently, the proportion of games with a positive adjusted streak statistic is  
593 correspondingly low, as reflected by the 4.3% figure.

594 In games with more than one opportunity, the proportion of games exhibiting a positive  
595 adjusted streak statistic is substantially larger than in single-opportunity games, with all  
596 but the 13-opportunity category exceeding 50%. This pattern can be explained by the  
597 underlying structure of the data, as multiple opportunities within a game typically arise  
598 from prior successful opportunity shots, leading to higher opportunity field goal percentages  
599 and, consequently, larger adjusted streak statistics. For instance, consider a shot sequence  
600 with five opportunity shots. Due to the limited number of shots that a player can take  
601 within a game, it is very likely that the opportunity shots clump together, forming a string of  
602 successful opportunity shots. The example shot sequences below illustrates this concept,  
603 with each containing five opportunity shots.

604 **Sequence 1:**

605 0, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 0

606 **Sequence 2:**

607 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0

### Sequence 3:

0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0

The difference in proportions between opportunity and non-opportunity shots for each sequence is given by:

$$[ \text{Sequence 1: } \frac{4}{5} - \frac{5}{10} = 0.3 ] [ \text{Sequence 2: } \frac{3}{5} - \frac{6}{10} = 0 ] [ \text{Sequence 3: } \frac{1}{5} - \frac{12}{20} = -0.4 ]$$

Sequences 1 and 2 contain opportunity shots that are more tightly clustered, necessitating a sequence of successful opportunity shots. This results in higher opportunity field goal percentages, which ultimately causes the streak statistic to increase. In contrast, sequence 3 contains opportunity shots that are more dispersed throughout the sequence, indicating more unsuccessful opportunity shots. This structure decreases the field goal percentage on opportunity shots, resulting in a lower streak statistic.

Importantly, sequences similar to 1 and 2 are more likely to occur in real game data, as the number of shots taken during the duration of a game are limited and multiple opportunities typically arise from consecutive made shots. This structural feature likely contributes to the increased number of games with positive bias-adjusted streak statistics for games with a large number of opportunity shots.



624

Chapter 5

625

LOGISTIC HIERARCHICAL MODELS

626 While the permutations tests for both season-long and within-game sequences provide little  
627 evidence of a hot-hand effect, these analyses do not account for contextual factors that may  
628 influence shooting performance. Incorporating additional variables into our analysis allows  
629 for a reexamination of the hot-hand effect while accounting for specific in-game conditions.

630 **5.1 Shot Distance Analysis**

631 Shot distances were analyzed to assess whether players exhibit different shooting distances  
632 on opportunity shots compared to non-opportunity shots. Summary statistics for quantitative  
633 shot distance by opportunity status are presented in Table 5.1, where “all” includes both  
634 opportunity and other shots.

Table 5.1: Average Shot Distance by Shot Type.

| Shot Type               | Average Distance (Ft.) |
|-------------------------|------------------------|
| All                     | 14.28                  |
| Opportunity             | 16.03                  |
| Other (non-opportunity) | 14.13                  |

635 The table reveals that opportunity shots are taken from a greater average distance than  
636 other shots, suggesting increased difficulty in these situations. Figure 5.1 displays the  
637 distribution of shot distances by shot type, with both curves exhibiting a similar bimodal  
638 shape corresponding to close-range and three-point attempts. However, while the peaks of  
639 the “other” shot distribution are of roughly equal height, the opportunity shot distribution

640 features a more pronounced peak at longer distances compared to its peak at shorter distances.  
 641 This shift indicates that opportunity shots are more likely to be taken from longer distances  
 642 than other shots, reinforcing the pattern observed in Table 5.1.

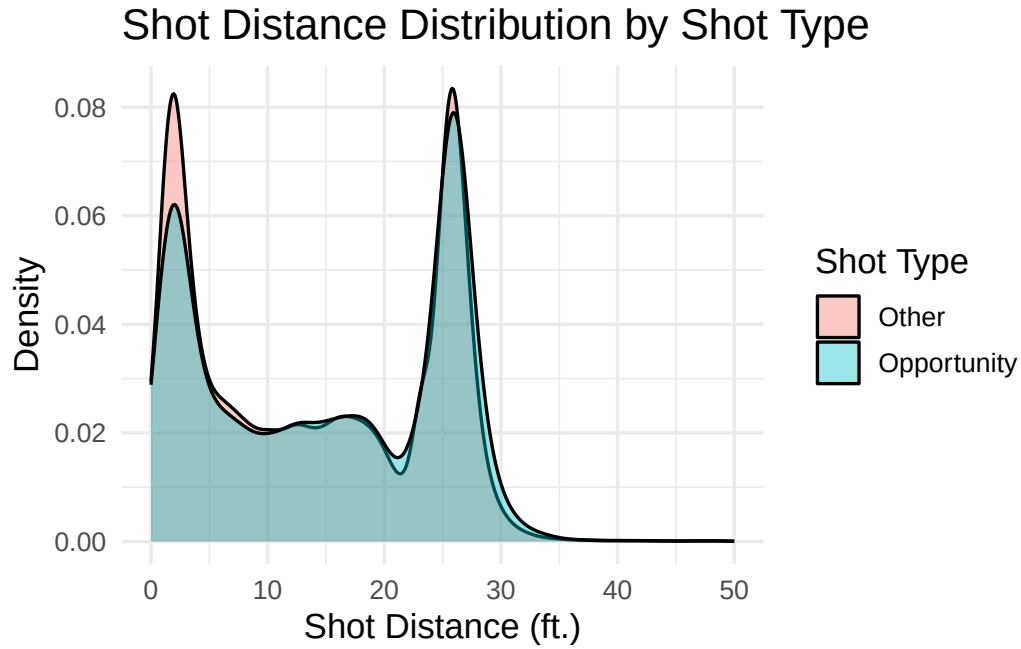


Figure 5.1: Shot Distances by Shot Type.

643 To further solidify the differences between opportunity and non-opportunity (other) shots,  
 644 the average shot distance for each shot type was computed within each player-season to  
 645 account for player-season variation. For each player-season, the difference between mean  
 646 opportunity shot distance and mean non-opportunity shot distance was calculated. The mean  
 647 of these differences was approximately 2.14 feet, indicating that, on average, opportunity  
 648 shots were taken from a greater distance than other shots within the same player-season.

649 A paired  $t$ -test was conducted to assess whether the population mean difference in shot  
 650 distance was greater than zero. The results provide strong evidence that opportunity shots  
 651 are taken from a greater distance than non-opportunity shots,  $t(499) = 30.94$ ,  $p < 0.001$ .  
 652 The estimated mean difference was 2.14 feet, with a one-sided 95% confidence interval  
 653 indicating that the true mean difference is at least 2.03 feet.

Figure 5.2 presents a scatterplot of average shot distance on opportunity shots versus average shot distance on non-opportunity shots for each player-season, grouped by position. Positions were categorized into guard, forward, and center based on classifications from Basketball Reference (“NBA League Averages - Per Game,” n.d.). The diagonal reference line represents equality in shot distance between the two shot types. The majority of points lie above this line, indicating that, for most player-seasons, opportunity shots are taken from greater distances than non-opportunity shots.

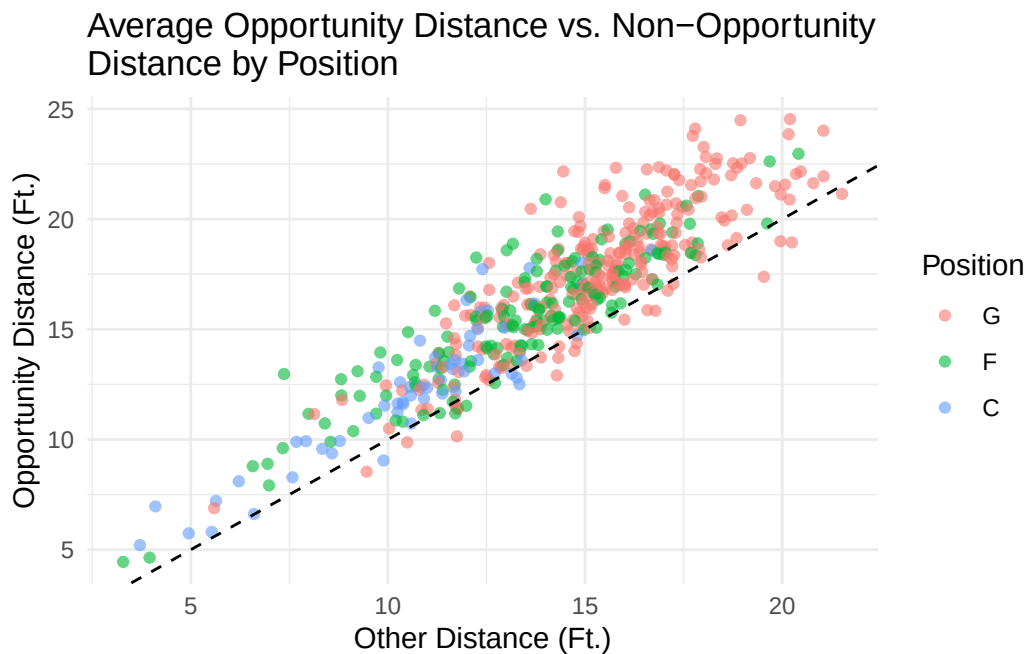


Figure 5.2: Average Shot Distances by Shot Type for Qualified Player-Seasons.

Shot distance was also analyzed on a categorical scale to account for the large proportion of shots taken from grouped distances. The three levels of shot distance are close (0-8 feet), mid-range (8-22 feet), and three-pointer (22+ feet), as outlined in Section 3.2.2.

Table 5.2: Proportions of Distance Category by Shot Type.

| Shot Type   | Close | Mid-range | Three-Pointer |
|-------------|-------|-----------|---------------|
| Opportunity | 0.333 | 0.286     | 0.380         |

| Shot Type | Close | Mid-range | Three-Pointer |
|-----------|-------|-----------|---------------|
| Other     | 0.376 | 0.283     | 0.341         |

Table 5.2, along with Figure 5.3, reveals that a larger proportion of opportunity shots were three-pointers compared to other shots, while a smaller proportion were close shots. The proportion of mid-range shots remained relatively similar between shot types. A chi-square test of independence confirmed that the distribution of distance categories is dependent on shot type  $\chi^2(2) = 420.55, p < .001$ .

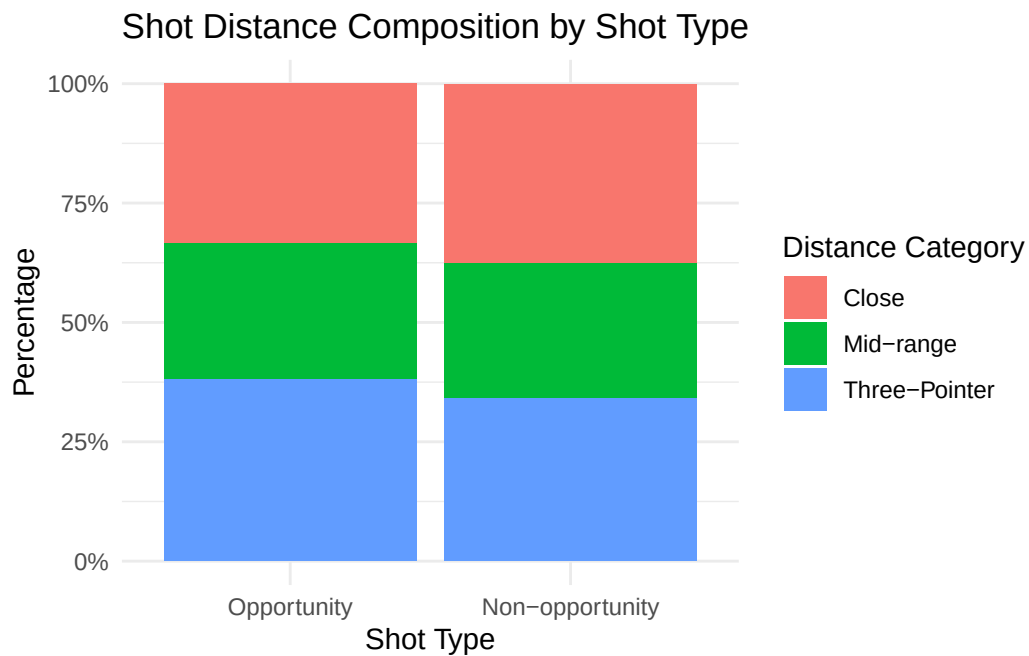


Figure 5.3: Distance Category by Shot Type.

This phenomenon likely arises from a combination of psychological and strategic factors. Players may describe the sensation of being “on fire”, referring to a mental and physical state of heightened focus and elevated performance. This can lead to increased confidence and manifest in more aggressive shot selection, including attempts from greater distances. The general public commonly refers to this state as heat-check behavior, as players attempt to

674 capitalize on their recent success. Other possible explanations involve strategic adjustments  
675 from both teams. Defenses may adjust by applying tighter coverage to a perceived “hot”  
676 player, forcing shots from further distances or worse positions. In turn, offenses may settle  
677 for longer-distance shots, as these are often quicker to attempt or more readily available  
678 under increased defensive pressure.

679 Due to this phenomenon, a new model that includes shot distance as a variable would greatly  
680 improve the results of the analysis by further isolating the hot-hand effect.

## 681 5.2 Model Specification

682 A logistic hierarchical model was constructed to evaluate the probability of a successful shot  
683 based on specific in-game variables and random effects. The variables and random effects  
684 are outlined in Table 5.3.

$$\begin{aligned}\text{logit}(y_{ijk}) &= \beta_0 + \beta_1 \text{opportunity}_{ij} + u_{0j} + u_{1j} \text{opportunity}_{ij} + \epsilon_{ij} \\ u_{0j} &\sim \mathcal{N}(0, \sigma_{u0}^2) \\ u_{1j} &\sim \mathcal{N}(0, \sigma_{u1}^2) \\ \epsilon_{ij} &\sim \mathcal{N}(0, \sigma^2)\end{aligned}$$

Table 5.3: Hierarchical Model Variable Descriptions.

| Variable       | Description                                                 | Effect Type |
|----------------|-------------------------------------------------------------|-------------|
| make           | Baseline shot distance                                      | Response    |
| intercept      | Indicator for whether a shot was made (1 = make, 0 = miss)  | Fixed       |
| is_opportunity | Indicator for opportunity shot (1 = yes, 0 = no)            | Fixed       |
| dist_category  | Categorical shot distance (Close, Mid-range, Three-Pointer) | Fixed       |

|                                |                                                      |        |
|--------------------------------|------------------------------------------------------|--------|
| is_opportunity*dist_category   | Interaction between is_opportunity and dist_category | Fixed  |
| (1 &#124; player)              | Player-specific intercept                            | Random |
| (is_opportunity &#124; player) | Player-specific opportunity effect                   | Random |
| (1 &#124; player)              | Season-specific intercept                            | Random |

---

Hot-hand theory posits that the probability of a successful shot increases on opportunity shots compared to non-opportunity shots, resulting in a positive coefficient for the opportunity variable. The inclusion of this variable allows us to assess how the probability of a successful shot varies with opportunity status while simultaneously accounting for other factors.

Player and season random effects were also included in the hierarchical model to account for unexplained heterogeneity in shooting performance and opportunity effects. In particular, the model includes random slopes for the opportunity effect across different players rather than assuming a uniform effect. Allowing different slopes for opportunity effects allows us to see which players are most impacted, or potentially not impacted, by hot-hand opportunities. Season random intercepts were also included to account for differences in baseline shooting ability across seasons.

### 5.3 Coefficient Bias

The opportunity coefficient is prone to the small-sample bias present in permutation testing. Therefore, the coefficient becomes inflated and overestimates the true effect of an opportunity shot on success probability. To account for this, a random sample of 100 games was selected from the full dataset. The following steps were taken to obtain a null distribution of opportunity coefficients from the sample:

1. Randomly permute the shot sequences within the 100 sampled player-games to simulate independent shots.
2. Fit the logistic hierarchical model for the simulated data to obtain a null opportunity

705 coefficient.

706 3. Repeat steps 1-2 many times (e.g., 10,000 trials) to obtain a null distribution of  
707 opportunity coefficients.

708 4. Calculate the mean of the null distribution to obtain the opportunity coefficient ex-  
709 pected under independence.

710 The procedure described above contained only 100 games due to computational limitations.

711 Therefore, the sample may not fully represent the full dataset, but should provide a general  
712 estimate for the opportunity coefficient expected under independence.

## 713 5.4 Results

714 The null distribution of opportunity coefficients for the 100 randomly sampled games is  
715 displayed in Figure 5.4. The mean of this distribution provides an estimate of the opportunity  
716 coefficient expected under independence and will be used to determine if the observed  
717 coefficient exceeds what would be expected under independence.

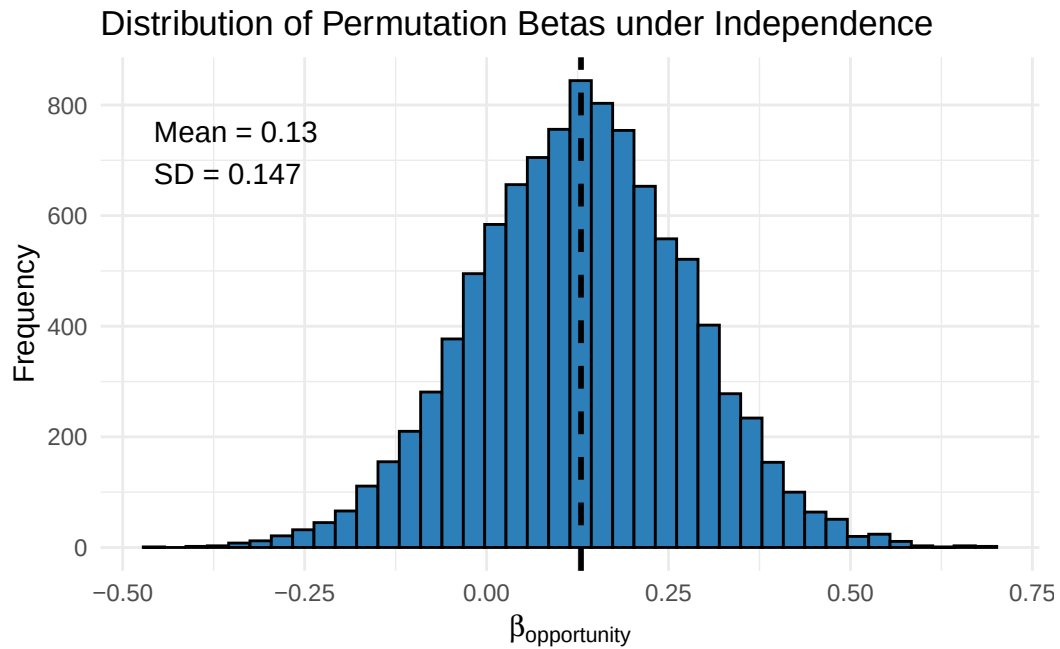


Figure 5.4: Histogram of Simulated Opportunity Coefficients under Independence.

718 The mean opportunity coefficient of the simulated trials under independence is approximately  
719 0.13, suggesting the existence of a bias that overestimates the true hot-hand effect. We now  
720 compare these results to the coefficients obtained from the observed hierarchical model.  
721 The fixed coefficient estimates for the full data are listed in Table 5.4 and random effect  
722 coefficients are listed in Table 5.5.

Table 5.4: Fixed Effect Coefficients.

| Coefficient                            | Estimate | Std. Error | z value   | p-value |
|----------------------------------------|----------|------------|-----------|---------|
| Intercept                              | 0.3207   | 0.0173     | 18.5423   | 0e+00   |
| is_opportunity                         | 0.0634   | 0.0168     | 3.7856    | 2e-04   |
| dist_categoryMidrange                  | -0.6126  | 0.0070     | -87.1207  | 0e+00   |
| dist_categoryThreePoint                | -0.8927  | 0.0068     | -130.3290 | 0e+00   |
| is_opportunity:dist_categoryMidrange   | -0.0826  | 0.0245     | -3.3723   | 7e-04   |
| is_opportunity:dist_categoryThreePoint | -0.1224  | 0.0232     | -5.2717   | 0e+00   |

Table 5.5: Random Effects (Standard Deviations)

| Group  | Effect         | Std. Dev. |
|--------|----------------|-----------|
| Player | Intercept      | 0.1056    |
| Player | is_opportunity | 0.0002    |
| Season | Intercept      | 0.0449    |

723 The opportunity indicator variable yields a coefficient of 0.0634, which is notably lower than  
724 the expected opportunity coefficient under independence. This suggests that the estimated  
725 hot-hand effect in the hierarchical model is actually weaker than what would be expected if  
726 shots were independent. The statistically significant  $p$ -value for the opportunity coefficient  
727 in Table 5.4 should be interpreted with caution, as it is derived by comparing the estimated



728 coefficient to 0. However, the simulation of null opportunity coefficients for the 100  
729 randomly sampled games yielded an estimated coefficient of 0.13, which represents the  
730 actual value that should be compared to our observed coefficient. Figure 5.5 displays the  
731 observed opportunity coefficient in relation to the null coefficients, along with the  $p$ -value  
732 describing the proportion of simulated trials with an opportunity effect as strong as the  
733 observed effect.

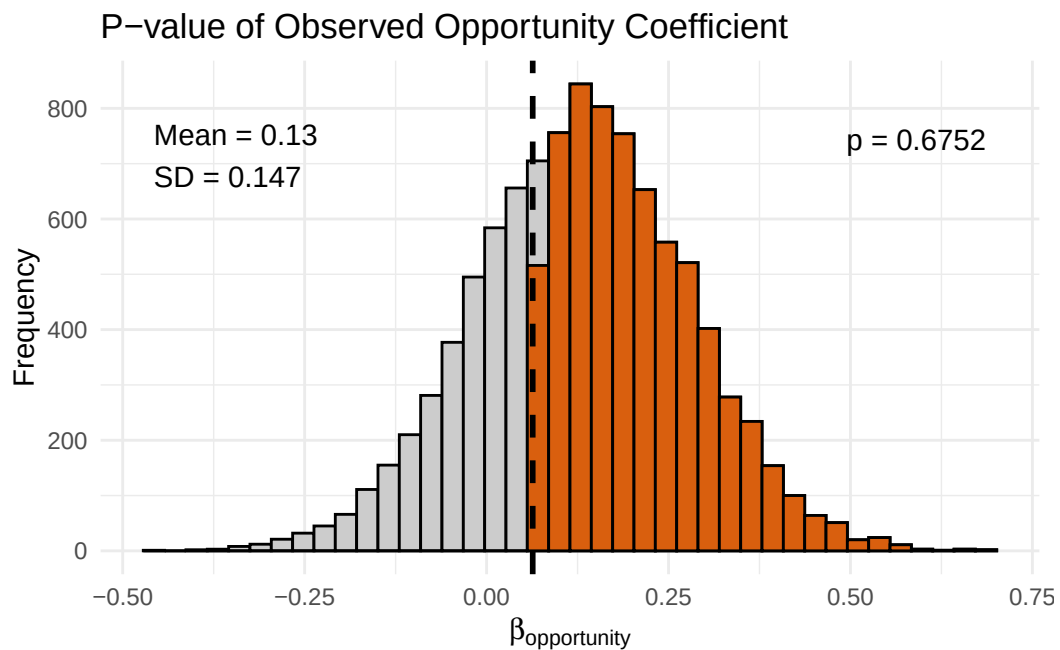


Figure 5.5: P-value of observed opportunity effect compared to null opportunity effects.

734 A large  $p$ -value of 0.6752 is obtained from the simulation, indicating weak evidence of  
735 an overall hot-hand effect over the past 10 seasons for high-usage players. The observed  
736 opportunity coefficient is consistent with what would be expected under the assumption of  
737 independence.

738 As expected, we also observe increasingly negative coefficients as shots are taken further  
739 from the basket. The mid-range and three-point indicators both show substantial decreases  
740 relative to close-range shots, reflecting the increase in shot difficulty with distance. The  
741 interaction between opportunity and distance is also negative for both mid-range and three-

742 point shots, indicating that the impact of an opportunity shot decreases as shot distance  
743 increases.

744 The player-level random intercept yields a relatively large standard deviation, representing  
745 the substantial heterogeneity in baseline shooting ability across players. This is expected in  
746 basketball data, as skillsets vary drastically across players and contribute to different baseline  
747 shooting percentages. In contrast, the standard deviation of random slopes for the opportunity  
748 indicator is extremely small, suggesting that the opportunity effect is relatively consistent  
749 across players. The season-level random intercept is non-negligible, indicating that the  
750 baseline shooting ability varies across seasons. This could reflect changes in offensive  
751 strategy, shooting ability, or other contextual factors that could impact shooting efficiency  
752 from year to year.

## 6.1 Limitations

Importantly, the results obtained from the permutation test are specific to the sample of players considered, which primarily consist of high-usage players who attempt a large number of shots for their respective teams. As such, these findings do not generalize to low-usage players, whose shot selection and opportunities may differ substantially. Additionally, the permutation testing framework does not account for contextual factors that may influence the perception or existence of a streak.

Within-game shot sequences also present a challenge for detecting hot-hand patterns due to limited sample sizes. The average number of shots per game for players in the sample was approximately 16, resulting in relatively few opportunity shots and, consequently, low statistical power. Considering season-long shot sequences increases the power of detecting a hot-hand by increasing sample sizes and increasing the number of opportunities. However, the results lose practical interpretability, as the hot-hand effect is generally understood to occur within games rather than across extended periods.

Additional limitations arise from modeling assumptions and measurement choices. The hierarchical model assumes independence between shots, which is partially addressed through the inclusion of the opportunity variable. However, the opportunity variables does not fully capture within-game dynamics that may induce dependence of shots. Furthermore, defining an opportunity as three consecutive made shots serves only as a proxy for a “hot” state and may not fully reflect the underlying hot-hand effect.

The bias-adjustment procedure in the hierarchical analysis used a sample of 100 games

776 to determine a null distribution of beta coefficients for the opportunity effect. While this  
777 procedure simulates independent shot sequences, the resulting null distribution is specific to  
778 the 100 randomly sampled games and may not fully capture the variability present in the  
779 complete dataset. As a result, the estimated bias adjustment may be sensitive to the particular  
780 sample of games selected, potentially limiting the generalizability of the findings.

## 781 **6.2 Future Work**

782 Additional analyses can be implemented to address the limitations presented above. Using  
783 a larger sample of players would allow for greater generalization to the NBA as a whole  
784 rather than only high-volume players. Incorporating additional variables, such as time  
785 between shots, defensive pressure, and game situation would help to further isolate the  
786 hot-hand effect and distinguish it from general shot streakiness. The permutation test and  
787 hierarchical model in this analysis employed a streak length of 3, which limits the frequency  
788 of opportunities. Shortening the streak length (e.g.,  $k = 1$ ) would increase the number of  
789 opportunities within a shot sequence and, therefore, statistical power, but could weaken the  
790 contextual interpretation of an opportunity as representing a genuinely “hot” state. Future  
791 research could also extend this analysis to incorporate temporal dependence and a deeper  
792 examination of what constitutes a “hot” state.

793 Increasing the number of sampled games would improve the bias adjustment for the hi-  
794 erarchical model coefficient by providing a more stable estimate of the null opportunity  
795 coefficients. Future analyses can look to construct hierarchical models on an individual  
796 player level to account for the variation between players. The bias adjustment would then  
797 pertain to specific players rather than the sample as a whole.

### 798 **6.3 Computational Details**

799 The results in this paper were obtained using R 4.5.3. R itself and all packages used are avail-  
800 able from the Comprehensive R Archive Network (CRAN) at <https://CRAN.R-project.org/>.

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